

Muhammad Mueed-Ul-Haq

SCADA / Industrial IoT Developer — Energy Industry

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Energy engineer with 6+ years across power electronics controls, grid simulation and battery storage, now focused on SCADA and Industrial IoT integration of distributed energy assets into E.ON's virtual power plant. I bridge the field level (inverters, PLCs, protocols) and the platform level (dispatch, monitoring, 24/7 operations).

CORE SKILLS

Industrial communication: Modbus TCP/RTU, IEC 60870-5-104, SCADA, Node-RED

Power electronics: grid-forming / grid-following controls, inverters, DC/DC

Data & AI: ML-based control, predictive maintenance

Energy systems: VPP, BESS, DER integration, grid codes, energy markets

Simulation & testing: MATLAB/Simulink, OpenDSS, PowerPanda, HIL, SIL

Leadership: technical lead in R&D, mentoring, patent filing

EXPERIENCE

Senior Energy Engineer — Asset Integration & Operation CURRENT

Oct 2025 – Present

E.ON · Essen, Germany

- Integrate battery storage and distributed energy assets into E.ON's virtual power plant, from SCADA interface design to commissioning and go-live.
- Implement asset communication over Modbus TCP and IEC 60870-5-104 and build integration flows on a Node-RED based stack.
- Develop battery steering logic that turns VPP schedules into setpoints within SoC, power and ramp-rate limits.
- Support 24/7 operation of the asset portfolio: monitoring, troubleshooting and root-cause analysis with OEMs.
- Standardise and automate asset onboarding to shorten time from contract to dispatchable asset.

Senior Power Electronics Controls Engineer

Apr 2025 – Sep 2025

Fluence (A Siemens and AES Company) · Power Conversion Systems · Erlangen

- Drove grid-forming control development for PCS in utility-scale energy storage.
- Designed and simulated control algorithms for inverters and DC/DC converters.
- Validated controls through prototyping, HIL and SIL testing; technical lead and mentor for junior engineers.

Power Electronics Engineer

Dec 2023 – Mar 2025

E.ON Group Innovation · R&D · Munich

- Led design and implementation of advanced BESS and grid performance strategies.
- Identified two technology gaps in BESS and grid-tied inverters and filed patents.
- Worked across energy markets, interconnection standards and regional grid codes.

Working Student, R&D

Nov 2022 – Nov 2023

E.ON Digital Technology · Munich

- Simulated IEEE test feeders (OpenDSS, PowerPanda, MATLAB) under faults and load variations; cross-validated results and reliability metrics.

EXPERIENCE (CONTINUED)

Working Student, Magnetics
May 2022 – Oct 2022

TDK Electronics · Munich

- Designed and tested magnetic components and common-mode chokes; lab measurements for datasheets.

Analyst, E-Mobility
Aug 2020 – Apr 2022

Power Technology Research · Lahore

- EV market research for Fortune 500 clients; articles, infographics and a talk at Enlit Europe 2021.

Research Associate
May 2020 – Aug 2020

BarqBox · Lahore

- IoT-enabled Li-ion batteries for light commercial vehicles with remote switching and monitoring.

SELECTED PROJECTS

BESS Integration into the Virtual Power Plant
E.ON · 2025 – now — SCADA interfaces and steering logic for battery assets.

Degradation-aware BESS Strategies
E.ON SE x TU Munich · 2024 – 2025 — led team; market bidding, AI predictive maintenance.

Analog Active Cell Balancing BMS
LUMS · 2019 – 2020 — bi-directional Ćuk converter, analog control.

Real-time Asset Telemetry
E.ON · 2025 – now — Modbus TCP / IEC 104 data flows for 24/7 operations.

ML-based Adaptive 3-Phase Inverter
E.ON SE x TU Munich · 2023 — grid-forming / grid-following control with ML.

EDUCATION

M.Sc. Power Engineering — Technical University of Munich
Oct 2021 – Nov 2023

Grade 1.8 · Power Electronics, Advanced Controls, Energy Storage, Renewable Integration

B.Sc. Electrical Engineering — LUMS, Lahore
Aug 2016 – May 2020

CGPA 3.63 / 4.0 (German grade 1.37) · Graduated with Distinction

PUBLICATIONS, TALKS & HONORS

IEEE ICDCM 2021 — First Author
Detachable Lithium-Ion Starter Battery and Analog BMS for ICE Vehicles.

Award of Distinction, LUMS
2020

Enlit Europe 2021, Milan — Speaker
Substation automation: centralised vs. distributed architectures.

Dean's Honor List, LUMS
2020